

18-427/727: Board-Level RF Systems for IoT

LAB06: Beam-Steering & LoRa

Writeup Due: Friday, December 5th, 2025

How to submit labs:

Create a word document or PDF that answers the questions presented below. Include any screenshots or figures as requested. **Please provide a descriptive caption for each figure!**

Question:	1	Total
Points:	100	100
Score:		

Lab Outline

In the previous lab, you learned about antenna arrays and beam-steering through simulation. In this lab, you will get to see this in practice through a physical implementation of a beam-steering system! The teaching staff will set up a receiver system using a KrakenSDR and a 5x1 whip antenna array. The radio will be listening to RF signals on the 915 MHz band. The lab groups will take turns broadcasting LoRa signals at different points in the classroom, and we will see how good of a job the Kraken does to track the transmitters! You'll also work through some analysis questions to develop a better understanding of beam-steering and LoRa.

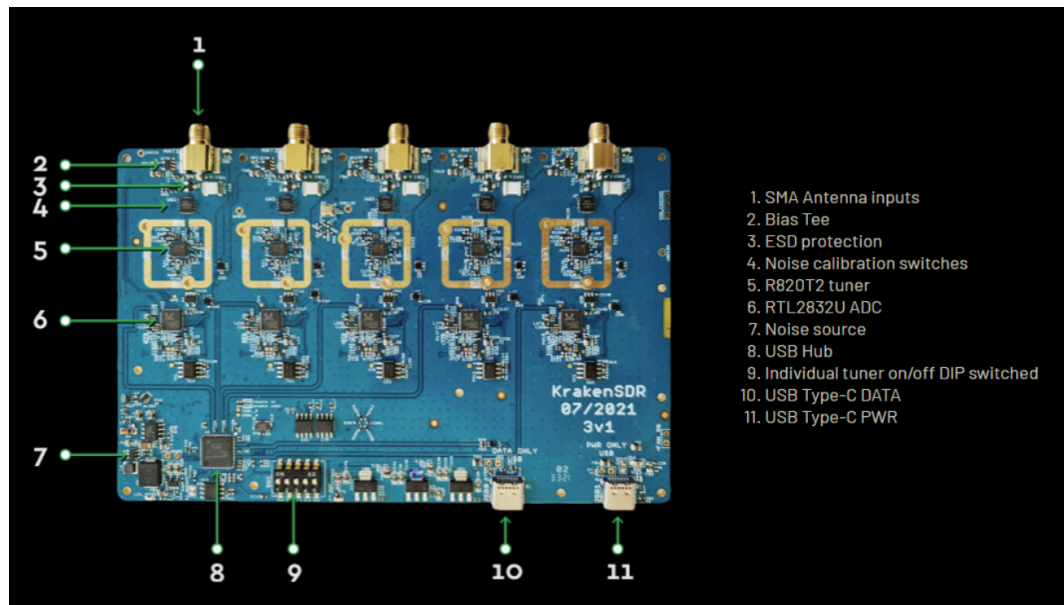
1. Setup
2. Demonstration
3. Analysis

Equipment Required

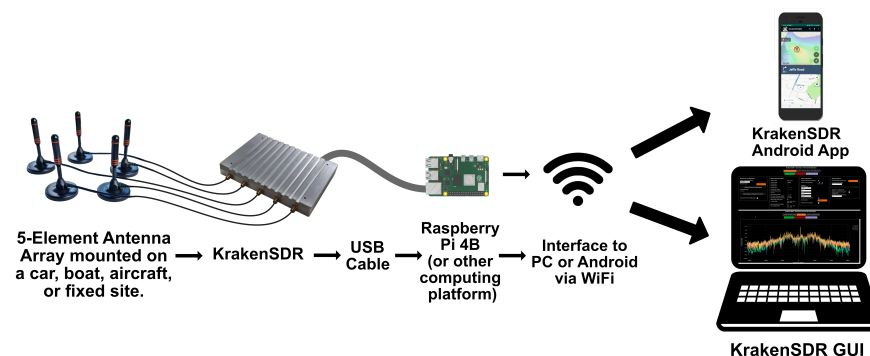
- Laptop - to power the LoRa microcontroller
- LoRa microcontrollers - to transmit LoRa signals to the receive station
- Patch antenna - to connect to your microcontroller for LoRa TX

1. Setup, Demonstration, and Analysis

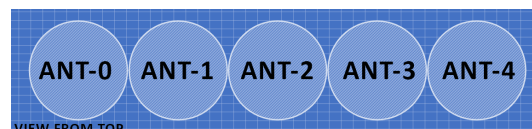
In this lab, the teaching staff will be using the KrakenRF to receive LoRa-modulated signals. The Kraken PCB is shown below:



This radio is effectively five RTL-SDRs on the same PCB, all synchronized to the same oscillator and clock signals. The Kraken is connected to a Raspberry Pi 4, which runs the software that interfaces with the Kraken to collect data. The Pi 4 is also connected to a wireless hotspot, which allows the device to stream the data live to a web interface for viewing.

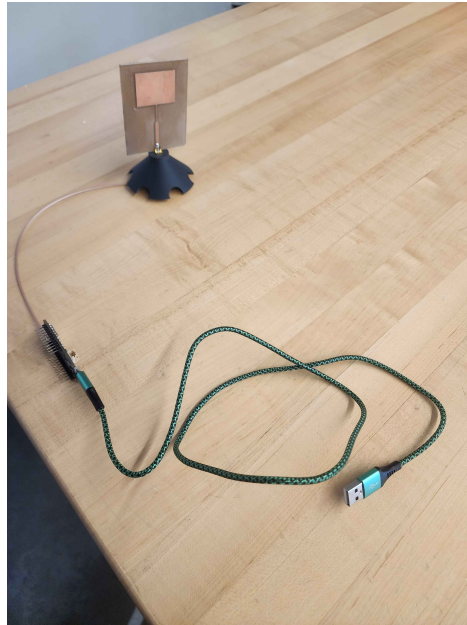


The five inputs to the Kraken are connected to a linear array of whip antennas as depicted below:



This array is on an aluminum sheet, which acts as a ground plane. The whip antennas are designed to operate at 915 MHz, which is a popular band for LoRa communications. This setup allows the Kraken to act as an RX beam-steering system, which can be used for radio direction finding. We'll talk more about the operating principles behind this implementation later on.

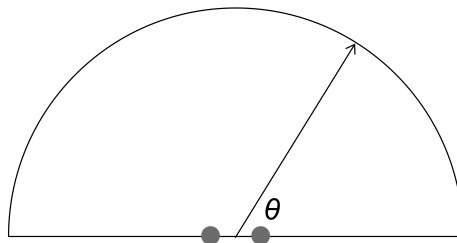
In this lab, each lab group will get to use an [Adafruit Feather M0 microcontroller](#), which has an RFM95 LoRa radio:



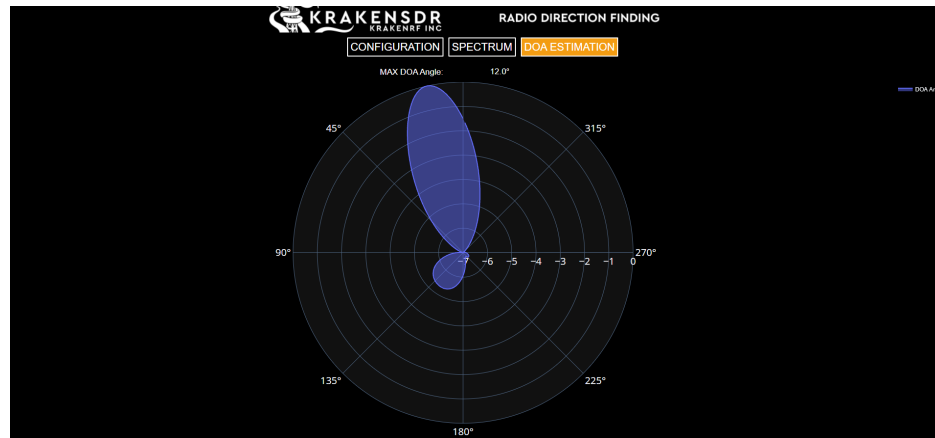
This device is capable of receiving and demodulating 868 MHz or 915 MHz signals, as well as transmitting signals in the same band. It is also capable of both LoRa and FSK modulation. These microcontrollers are already programmed with CircuitPython; take a moment to read over [this page](#), which explains how the microcontroller works in more detail. Most of the action happens in the hardware, which is proprietary, so there isn't too much to see in the software. You can find the code that is running on the microcontroller in the Lab 6 folder, if you're curious to see what's happening!

Once you connect your microcontroller to your laptop, it will automatically be powered and start sending LoRa chirps. Be sure to connect the patch antenna to the microcontroller. It uses a U.FL connector, which has a form factor that makes it convenient to fit onto smaller devices. However, it can be a bit difficult to ensure that the antenna stays connected. You should feel a soft click when the antenna is connected; try not to move it around too much, or it will disconnect from the microcontroller.

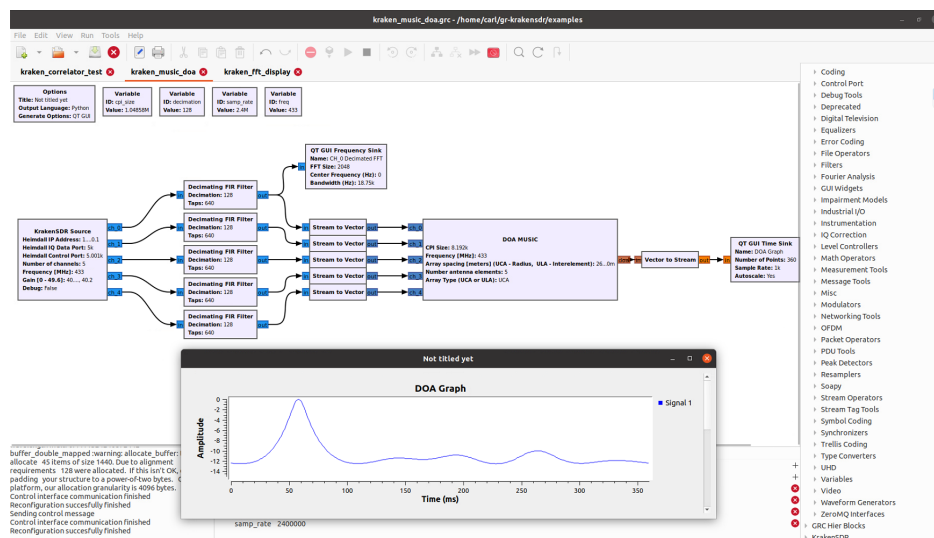
You will be seated in a semicircle for this final demonstration as pictured below:



The teaching team has configured the KrakenSDR to receive signals at 915 MHz, and they will have the web interface pulled up on the display:



There are many ways that one can implement direction finding to hone in on the location of a radio transmitter. The [KrakenRF wiki](#) provides some background in the theory behind direction finding; the KrakenRF specifically uses correlative interferometry, which looks at the phase between received signals in the antenna array. The wiki also provides various [GNU Radio models](#) that show the capabilities of the Kraken radio, such as the following direction-of-arrival block diagram that relies upon the [MUSIC \(MUltiple SIGNAL Classification\)](#) algorithm:



We will have two to four lab groups transmit packets at a time, so that you can see the different lobes on the web interface. The teaching team will ensure that the groups that are transmitting are a few meters from each other so that we can more easily see clearly defined lobes.

Now, please answer the following questions to help reinforce your understanding of LoRa and beam-steering:

10 pts

1.1 What is the approximate angle that you were seated at in the semicircle relative to the Kraken antenna array? What should the phase difference ψ be between the two elements, in degrees, that would result in the beam pointing in your direction? Keep in mind that this is a five-element array with element spacing $d = \frac{\lambda}{2}$.

10 pts

1.2 How far apart should each of the receive array's antennas be for operation in the 915 MHz band? What might happen if you spaced the antenna elements too far from each other?

10 pts

1.3 How does the spreading factor affect the data rate for LoRa signals? When would you want a lower SF?

10 pts

1.4 LoRa allows you to have an **implicit header**, as well as a **cyclic redundancy check (CRC)**. When would it be appropriate to not use these features?

10 pts

1.5 Explain how the MUSIC (multiple signal classification) algorithm being run on the Raspberry Pi works to process the data coming from the KrakenSDR.

10 pts

1.6 What is preventing you from taking five RTL-SDRs and connecting them to a computer for RX beam-steering? Why would this not yield the same results as the KrakenSDR?

10 pts

1.7 In addition to a linear array configuration, the Kraken allows you to configure it into a circular antenna array. When might this configuration be preferred?

10 pts

1.8 Why is the Kraken only able to track up to four targets at any given time?

10 pts

1.9 How might you deal with multipath propagation when you're trying to use the Kraken to locate the direction of a transmitter? (Hint: assume you have multiple Kraken radios available to you.)

10 pts

1.10 What other changes could we make to our setup (transmitters or receiver) to achieve better beam-steering results? This can be in the physical hardware or the software implementations.

Reflections

Please complete our [post-lab survey](#) for 5 points of extra credit! Attach a photo of the form completion page to your lab submission.